

hiv_gdp_analysis

May 13, 2026

1 HIV/AIDS Deaths & GDP Per Capita Analysis

1.1 1. Load raw dataframes

```
[34]: import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.cm as cm
import matplotlib.colors as mcolors
import numpy as np
import scipy.stats

aids_df = pd.read_csv('aids.csv')
aids_df['location'] = aids_df['location'].replace('North America', 'North_America - WB') # Named odd in original csv
gdp_df = pd.read_csv('gdpc.csv')

print('aids_df:', aids_df.shape)
display(aids_df.head(3))

print('\ngdp_df:', gdp_df.shape)
display(gdp_df.head(3))
```

aids_df: (56760, 11)

	population_group	measure	location	sex	age	\
0	All Population	Deaths	Sub-Saharan Africa - WB	Both	All ages	
1	All Population	Deaths	Sub-Saharan Africa - WB	Both	All ages	
2	All Population	Deaths	Sub-Saharan Africa - WB	Both	All ages	

	cause	metric	year	val	\
0	HIV/AIDS and sexually transmitted infections	Number	1980	42542.943666	
1	HIV/AIDS and sexually transmitted infections	Percent	1980	0.006862	
2	HIV/AIDS and sexually transmitted infections	Rate	1980	11.003186	

	upper	lower
0	76399.602417	18954.663455
1	0.012307	0.003075
2	19.759776	4.902380

gdp_df: (266, 70)

	Country Name	Country Code	Indicator Name	\
0	Aruba	ABW	GDP per capita (current US\$)	
1	Africa Eastern and Southern	AFE	GDP per capita (current US\$)	
2	Afghanistan	AFG	GDP per capita (current US\$)	

	Indicator Code	1960	1961	1962	1963	1964	\
0	NY.GDP.PCAP.CD	NaN	NaN	NaN	NaN	NaN	
1	NY.GDP.PCAP.CD	186.089204	186.909053	197.367547	225.400079	208.962717	
2	NY.GDP.PCAP.CD	NaN	NaN	NaN	NaN	NaN	

	1965	...	2016	2017	2018	2019	\
0	NaN	...	27441.550214	28440.041688	30082.158423	30645.890602	
1	226.836135	...	1334.470500	1528.104224	1552.073722	1507.085600	
2	NaN	...	522.082216	525.469771	491.337221	496.602504	

	2020	2021	2022	2023	2024	2025
0	22759.807175	26749.329609	30975.998912	35718.753119	39498.594129	NaN
1	1351.591669	1562.416175	1679.327622	1571.449189	1615.396356	NaN
2	510.787063	356.496214	357.261153	413.757895	NaN	NaN

[3 rows x 70 columns]

1.2 2. Reshape data

```
[35]: year_cols = [c for c in gdp_df.columns if str(c).isdigit()]

gdp_long = gdp_df.melt(
    id_vars=['Country Name', 'Country Code', 'Indicator Name', 'Indicator_
↳ Code'],
    value_vars=year_cols,
    var_name='year',
    value_name='gdp_per_capita'
)
gdp_long['year'] = gdp_long['year'].astype(int)

print('gdp_long:', gdp_long.shape)
display(gdp_long.head(3))
```

gdp_long: (17556, 6)

	Country Name	Country Code	Indicator Name	\
0	Aruba	ABW	GDP per capita (current US\$)	
1	Africa Eastern and Southern	AFE	GDP per capita (current US\$)	
2	Afghanistan	AFG	GDP per capita (current US\$)	

Indicator Code	year	gdp_per_capita
----------------	------	----------------

```

0 NY.GDP.PCAP.CD 1960 NaN
1 NY.GDP.PCAP.CD 1960 186.089204
2 NY.GDP.PCAP.CD 1960 NaN

```

1.3 3. Country name cleaning

```

[36]: name_map = {
    'Bahamas': 'Bahamas, The',
    'Bolivia (Plurinational State of)': 'Bolivia',
    'Congo': 'Congo, Rep.',
    'Democratic Republic of the Congo': 'Congo, Dem. Rep.',
    'Côte d'Ivoire': 'Cote d'Ivoire',
    'Egypt': 'Egypt, Arab Rep.',
    'Gambia': 'Gambia, The',
    'Iran (Islamic Republic of)': 'Iran, Islamic Rep.',
    'Kyrgyzstan': 'Kyrgyz Republic',
    'Lao People's Democratic Republic': 'Lao PDR',
    'Micronesia (Federated States of)': 'Micronesia, Fed. Sts.',
    'Republic of Korea': 'Korea, Rep.',
    'Democratic People's Republic of Korea': 'Korea, Dem. People's Rep.',
    'Republic of Moldova': 'Moldova',
    'Slovakia': 'Slovak Republic',
    'Somalia': 'Somalia, Fed. Rep.',
}

wb_region_map = {
    'Sub-Saharan Africa - WB': 'Sub-Saharan Africa',
    'Middle East & North Africa - WB': 'Middle East, North Africa, Afghanistan, & Pakistan',
    'South Asia - WB': 'South Asia',
    'East Asia & Pacific - WB': 'East Asia & Pacific',
    'Europe & Central Asia - WB': 'Europe & Central Asia',
    'Latin America & Caribbean - WB': 'Latin America & Caribbean',
    'North America - WB': 'North America'
}

```

1.4 4. Filter AIDS data to HIV/AIDS deaths only

```

[37]: # Remove other STIs
hiv_deaths = aids_df[aids_df['cause'] == 'HIV/AIDS'].copy()

if 'Number' in hiv_deaths['metric'].unique():
    hiv_deaths = hiv_deaths[hiv_deaths['metric'] == 'Rate']

print('Rows after filter:', len(hiv_deaths))
print('Metrics present: ', hiv_deaths['metric'].unique().tolist())
display(hiv_deaths.head(10))

```

Rows after filter: 9460

Metrics present: ['Rate']

	population_group	measure	location	sex	age	cause	\
8	All Population	Deaths	China	Both	All ages	HIV/AIDS	
23	All Population	Deaths	Equatorial Guinea	Both	All ages	HIV/AIDS	
26	All Population	Deaths	Türkiye	Both	All ages	HIV/AIDS	
29	All Population	Deaths	Uganda	Both	All ages	HIV/AIDS	
35	All Population	Deaths	Tajikistan	Both	All ages	HIV/AIDS	
41	All Population	Deaths	Belarus	Both	All ages	HIV/AIDS	
47	All Population	Deaths	Gabon	Both	All ages	HIV/AIDS	
50	All Population	Deaths	Algeria	Both	All ages	HIV/AIDS	
62	All Population	Deaths	Andorra	Both	All ages	HIV/AIDS	
68	All Population	Deaths	Guinea	Both	All ages	HIV/AIDS	

	metric	year	val	upper	lower
8	Rate	1980	0.000000	0.000000	0.000000
23	Rate	1980	0.030503	0.066116	0.004536
26	Rate	1980	0.020618	0.024645	0.016797
29	Rate	1980	13.310572	38.876086	2.390701
35	Rate	1980	0.635892	0.772814	0.509861
41	Rate	1980	0.399903	0.451599	0.346200
47	Rate	1980	0.059477	0.128317	0.011473
50	Rate	1980	0.047242	0.070896	0.026366
62	Rate	1980	0.031862	0.058645	0.014689
68	Rate	1980	0.031557	0.092546	0.003994

1.5 5. Build per-country joined dataframe

```
[38]: hiv_countries = hiv_deaths[~hiv_deaths['location'].str.endswith('- WB')].copy()

hiv_countries['location_gdp'] = hiv_countries['location'].map(name_map).
↳ fillna(hiv_countries['location'])

country_joined = hiv_countries.merge(
    gdp_long[['Country Name', 'year', 'gdp_per_capita']],
    left_on=['location_gdp', 'year'],
    right_on=['Country Name', 'year'],
    how='inner'
)

country_joined = country_joined.rename(columns={'location': 'country', 'val': 'hiv_aids_deaths'})
↳ hiv_aids_deaths

country_joined = country_joined[
    ['country', 'year', 'sex', 'age', 'population_group', 'measure',
     'hiv_aids_deaths', 'upper', 'lower', 'gdp_per_capita']
]
```

```

country_joined = country_joined.dropna(subset=['gdp_per_capita'])
print('country_joined:', country_joined.shape)
print(f'Countries : {country_joined["country"].nunique()}')
print(f'Years      : {country_joined["year"].min()} - {country_joined["year"].max()}')
display(country_joined.head(10))

```

country_joined: (7800, 10)

Countries : 188

Years : 1980 - 2023

	country	year	sex	age	population_group	measure \
0	China	1980	Both	All ages	All Population	Deaths
1	Equatorial Guinea	1980	Both	All ages	All Population	Deaths
2	Uganda	1980	Both	All ages	All Population	Deaths
5	Gabon	1980	Both	All ages	All Population	Deaths
6	Algeria	1980	Both	All ages	All Population	Deaths
7	Andorra	1980	Both	All ages	All Population	Deaths
8	Guinea	1980	Both	All ages	All Population	Deaths
10	Fiji	1980	Both	All ages	All Population	Deaths
11	Zambia	1980	Both	All ages	All Population	Deaths
12	United Arab Emirates	1980	Both	All ages	All Population	Deaths

	hiv_aids_deaths	upper	lower	gdp_per_capita
0	0.000000	0.000000	0.000000	195.149480
1	0.030503	0.066116	0.004536	175.451008
2	13.310572	38.876086	2.390701	93.750699
5	0.059477	0.128317	0.011473	5718.467191
6	0.047242	0.070896	0.026366	2275.779712
7	0.031862	0.058645	0.014689	12474.925292
8	0.031557	0.092546	0.003994	1930.991843
10	0.695059	0.908690	0.515376	1873.310980
11	0.732203	2.429929	0.054855	669.419722
12	0.048136	0.120503	0.015208	42925.233878

1.6 6. Build per-WB-region joined dataframe

```

[39]: hiv_regions = hiv_deaths[hiv_deaths['location'].str.endswith('- WB')].copy()
hiv_regions['location_gdp'] = hiv_regions['location'].map(wb_region_map)

hiv_regions = hiv_regions[hiv_regions['location_gdp'].notna()]

region_joined = hiv_regions.merge(
    gdp_long[['Country Name', 'year', 'gdp_per_capita']],
    left_on=['location_gdp', 'year'],
    right_on=['Country Name', 'year'],
    how='inner'
)

```

```

region_joined = region_joined.rename(columns={'location': 'wb_region', 'val': 'hiv_aids_deaths'})
region_joined = region_joined[
    ['wb_region', 'year', 'sex', 'age', 'population_group', 'measure',
     'hiv_aids_deaths', 'upper', 'lower', 'gdp_per_capita']
]

print('region_joined:', region_joined.shape)
print('Regions:', region_joined['wb_region'].unique().tolist())
print(f'Years : {region_joined["year"].min()} - {region_joined["year"].max()}')
display(region_joined.head(10))

```

region_joined: (308, 10)

Regions: ['Middle East & North Africa - WB', 'South Asia - WB', 'East Asia & Pacific - WB', 'Europe & Central Asia - WB', 'Latin America & Caribbean - WB', 'Sub-Saharan Africa - WB', 'North America - WB']

Years : 1980 - 2023

	wb_region	year	sex	age	population_group	\
0	Middle East & North Africa - WB	1980	Both	All ages	All Population	
1	South Asia - WB	1980	Both	All ages	All Population	
2	East Asia & Pacific - WB	1980	Both	All ages	All Population	
3	Europe & Central Asia - WB	1980	Both	All ages	All Population	
4	Latin America & Caribbean - WB	1980	Both	All ages	All Population	
5	Sub-Saharan Africa - WB	1980	Both	All ages	All Population	
6	North America - WB	1980	Both	All ages	All Population	
7	Latin America & Caribbean - WB	1981	Both	All ages	All Population	
8	Sub-Saharan Africa - WB	1981	Both	All ages	All Population	
9	North America - WB	1981	Both	All ages	All Population	

	measure	hiv_aids_deaths	upper	lower	gdp_per_capita
0	Deaths	0.066222	0.079593	0.054837	2223.902748
1	Deaths	0.002230	0.003399	0.001405	261.180796
2	Deaths	0.250912	0.343556	0.160373	1188.178471
3	Deaths	0.515104	0.534691	0.493444	5795.168203
4	Deaths	0.973975	0.995521	0.950935	2177.728865
5	Deaths	0.613272	1.476404	0.224678	769.565372
6	Deaths	0.980962	1.016271	0.948613	12441.490408
7	Deaths	0.951715	0.973721	0.928359	2434.064705
8	Deaths	1.167507	2.566418	0.492193	1007.860682
9	Deaths	0.972473	1.002646	0.940075	13820.193281

```

[40]: plot_df = region_joined[
    (region_joined['sex'] == 'Both') &
    (region_joined['age'] == 'All ages') &
    (region_joined['population_group'] == 'All Population') &
    (region_joined['wb_region'] != 'Sub-Saharan Africa - WB')

```

```

].copy()

years = plot_df['year'].values
year_min, year_max = years.min(), years.max()

cmap = cm.plasma
norm = mcolors.Normalize(vmin=year_min, vmax=year_max)
colors = cmap(norm(years))

regions = plot_df['wb_region'].unique()
markers = ['o', 's', '^', 'D', 'P', '*']          # one distinct marker per
region
region_marker = {r: markers[i % len(markers)] for i, r in enumerate(regions)}

fig, ax = plt.subplots(figsize=(11, 7))

for region, grp in plot_df.groupby('wb_region'):
    grp = grp.sort_values('year')
    region_colors = cmap(norm(grp['year'].values))
    sc = ax.scatter(
        grp['gdp_per_capita'],
        grp['hiv_aids_deaths'],
        c=grp['year'],
        cmap=cmap,
        norm=norm,
        marker=region_marker[region],
        s=70,
        edgecolors='white',
        linewidths=0.4,
        label=region,
        zorder=3
    )

# Colorbar for year
sm = cm.ScalarMappable(cmap=cmap, norm=norm)
sm.set_array([])
cbar = fig.colorbar(sm, ax=ax, pad=0.02)
cbar.set_label('Year', fontsize=11)
cbar.set_ticks(np.arange(year_min, year_max + 1, 5))

# Legend for region markers
ax.legend(
    title='WB Region',
    title_fontsize=9,
    fontsize=8,
    loc='upper left',
    framealpha=0.85

```

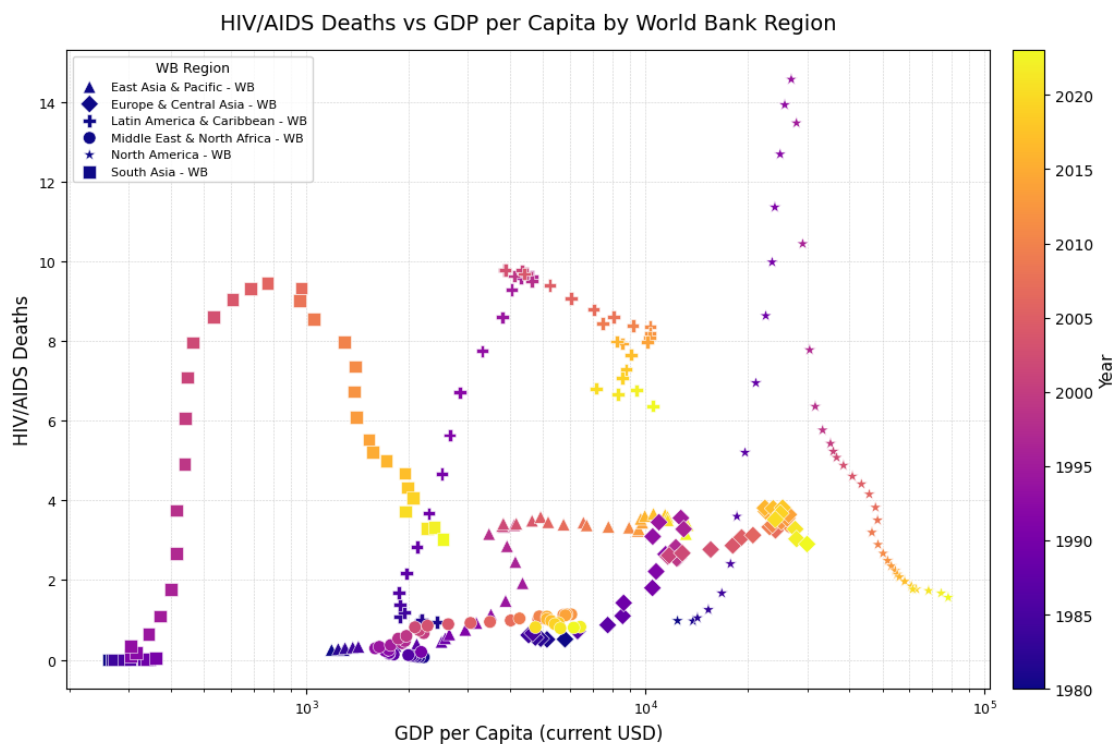
```

)

ax.set_xlabel('GDP per Capita (current USD)', fontsize=12)
ax.set_ylabel('HIV/AIDS Deaths', fontsize=12)
ax.set_title('HIV/AIDS Deaths vs GDP per Capita by World Bank Region',
             ↪ fontsize=14, pad=14)
ax.set_xscale('log')
ax.grid(True, which='both', linestyle='--', linewidth=0.4, alpha=0.6)
ax.tick_params(labelsize=9)

plt.tight_layout()
plt.show()

```



```

[41]: plot_df = country_joined[
        (country_joined['sex'] == 'Both') &
        (country_joined['age'] == 'All ages') &
        (country_joined['population_group'] == 'All Population')
    ].copy()

years = plot_df['year'].values
year_min, year_max = years.min(), years.max()

cmap = cm.plasma

```

```

norm = mcolors.Normalize(vmin=year_min, vmax=year_max)
colors = cmap(norm(years))

regions = plot_df['country'].unique()
markers = ['o', 's', '^', 'D', 'P', '*', 'X']
region_marker = {r: markers[i % len(markers)] for i, r in enumerate(regions)}

fig, ax = plt.subplots(figsize=(11, 7))

for region, grp in plot_df.groupby('country'):
    grp = grp.sort_values('year')
    region_colors = cmap(norm(grp['year'].values))
    sc = ax.scatter(
        grp['gdp_per_capita'],
        grp['hiv_aids_deaths'],
        c=grp['year'],
        cmap=cmap,
        norm=norm,
        marker=region_marker[region],
        s=70,
        edgecolors='white',
        linewidths=0.4,
        label=region,
        zorder=3
    )

# Colorbar for year
sm = cm.ScalarMappable(cmap=cmap, norm=norm)
sm.set_array([])
cbar = fig.colorbar(sm, ax=ax, pad=0.02)
cbar.set_label('Year', fontsize=11)
cbar.set_ticks(np.arange(year_min, year_max + 1, 5))

'''
ax.legend(
    title='WB Region',
    title_fontsize=9,
    fontsize=8,
    loc='upper left',
    framealpha=0.85
)
'''

ax.set_xlabel('GDP per Capita (current USD)', fontsize=12)
ax.set_ylabel('HIV/AIDS Deaths', fontsize=12)
ax.set_title('HIV/AIDS Deaths vs GDP per Capita by World Bank Region',
             ↵fontsize=14, pad=14)

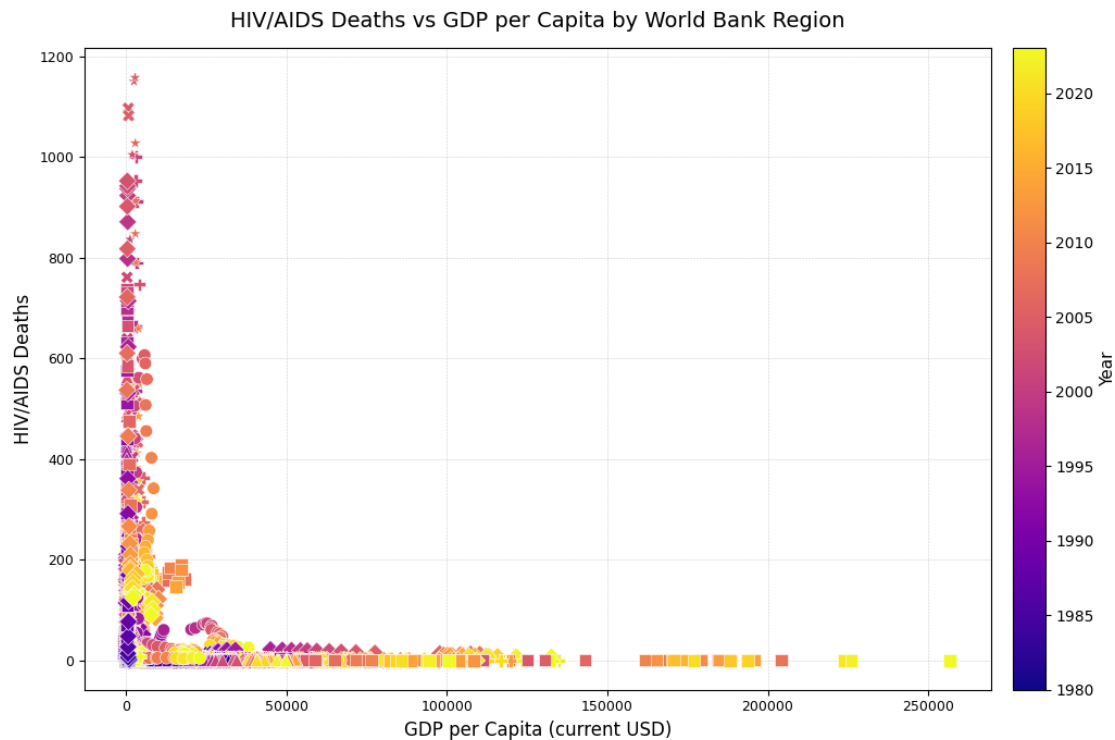
```

```

#ax.set_xscale('log')
ax.grid(True, which='both', linestyle='--', linewidth=0.4, alpha=0.6)
ax.tick_params(labelsize=9)

plt.tight_layout()
plt.show()

```



```

[42]: regions_lookup = pd.read_csv('world-regions-according-to-the-world-bank.csv')[
    ['Entity', 'World regions according to WB']]
    .drop_duplicates()

country_to_entity_map = {
    'Türkiye': 'Turkey',
    'Viet Nam': 'Vietnam',
    'Republic of Korea': 'South Korea',
    "Democratic People's Republic of Korea": 'North Korea',
    'Republic of Moldova': 'Moldova',
    'United Republic of Tanzania': 'Tanzania',
    'Venezuela (Bolivarian Republic of)': 'Venezuela',
    'Bolivia (Plurinational State of)': 'Bolivia',
    'Democratic Republic of the Congo': 'Democratic Republic of_
↳Congo',
    'Iran (Islamic Republic of)': 'Iran',

```

```

        'Syrian Arab Republic': 'Syria',
        'United States of America': 'United States',
        'Lao People's Democratic Republic': 'Lao PDR',
        'Brunei Darussalam': 'Brunei',
        'Côte d'Ivoire': 'Cote d'Ivoire',
        'Cabo Verde': 'Cape Verde',
    }

    plot_country_df = country_joined[
        (country_joined['sex'] == 'Both') &
        (country_joined['age'] == 'All ages') &
        (country_joined['population_group'] == 'All Population')
    ].copy()

    plot_country_df['entity'] = plot_country_df['country'].
        ↪map(country_to_entity_map).fillna(plot_country_df['country'])

    plot_country_df = plot_country_df.merge(
        regions_lookup,
        left_on='entity',
        right_on='Entity',
        how='left'
    )

    plot_country_df = plot_country_df.rename(columns={'World regions according to_
        ↪WB': 'wb_region'})

    plot_country_df = plot_country_df[
        #(plot_country_df['wb_region'] == 'Sub-Saharan Africa (WB)') &
        (plot_country_df['hiv_aids_deaths'] != 0.0)
    ].copy()

    missing = plot_country_df[plot_country_df['wb_region'].isna()]['country'].
        ↪unique()

    years = plot_country_df['year'].values
    year_min, year_max = years.min(), years.max()

    cmap = cm.plasma
    norm = mcolors.Normalize(vmin=year_min, vmax=year_max)

    regions = sorted(plot_country_df['wb_region'].dropna().unique())
    markers = ['o', 's', '^', 'D', 'P', '*', 'X']
    region_marker = {r: markers[i % len(markers)] for i, r in enumerate(regions)}

    fig, ax = plt.subplots(figsize=(12, 8))

```

```

for region, grp in plot_country_df.groupby('wb_region'):
    grp = grp.sort_values('year')
    ax.scatter(
        grp['gdp_per_capita'],
        grp['hiv_aids_deaths'],
        c=grp['year'],
        cmap=cmap,
        norm=norm,
        marker=region_marker[region],
        s=50,
        edgecolors='white',
        linewidths=0.3,
        label=region,
        zorder=3
    )

reg_df = plot_country_df[['gdp_per_capita', 'hiv_aids_deaths']].dropna()
reg_df = reg_df[reg_df['gdp_per_capita'] > 0] # guard against log(0)

x_all = np.log(reg_df['gdp_per_capita'].values)
y_all = np.log(reg_df['hiv_aids_deaths'].values)

slope, intercept, r_value, p_value, std_err = scipy.stats.linregress(x_all,
    ↪ y_all)
r_squared = r_value ** 2

x_log_line = np.linspace(x_all.min(), x_all.max(), 300)
y_line = np.exp(slope * x_log_line + intercept)
x_line = np.exp(x_log_line)

ax.plot(
    x_line, y_line,
    color='crimson',
    linewidth=1.8,
    linestyle='--',
    zorder=4,
    label=f'Linear regression on log(GDP) ( $R^2 = \{r\_squared:.3f\}$ )'
)

# Colorbar for year
sm = cm.ScalarMappable(cmap=cmap, norm=norm)
sm.set_array([])
cbar = fig.colorbar(sm, ax=ax, pad=0.02)
cbar.set_label('Year', fontsize=11)
cbar.set_ticks(np.arange(year_min, year_max + 1, 5))

ax.legend(

```

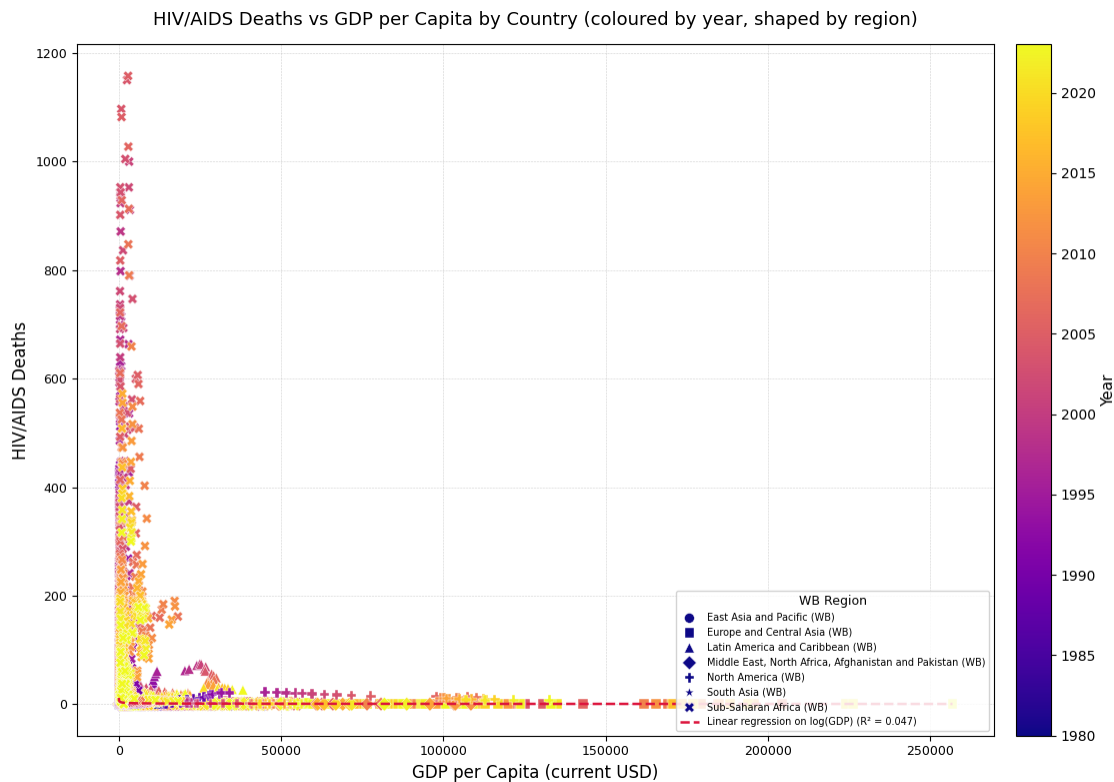
```

    title='WB Region',
    title_fontsize=9,
    fontsize=7,
    loc='lower right',
    framealpha=0.85
)

ax.set_xlabel('GDP per Capita (current USD)', fontsize=12)
ax.set_ylabel('HIV/AIDS Deaths', fontsize=12)
ax.set_title('HIV/AIDS Deaths vs GDP per Capita by Country (coloured by year, \
↳shaped by region)', fontsize=13, pad=14)
#ax.set_xscale('log')
#ax.set_yscale('log')
ax.grid(True, which='both', linestyle='--', linewidth=0.4, alpha=0.6)
ax.tick_params(labelsize=9)

plt.tight_layout()
plt.show()

```



```

[43]: import math

years_sorted = sorted(plot_country_df['year'].unique())

```

```

n_years = len(years_sorted)

# Grid layout
ncols = 5
nrows = math.ceil(n_years / ncols)

fig, axes = plt.subplots(nrows, ncols, figsize=(ncols * 4, nrows * 3.5),
    constrained_layout=True)
axes_flat = axes.flatten()

r2OverTime = []

for ax_idx, year in enumerate(years_sorted):
    ax = axes_flat[ax_idx]
    yr_df = plot_country_df[plot_country_df['year'] == year]

    for region, grp in yr_df.groupby('wb_region'):
        ax.scatter(
            grp['gdp_per_capita'],
            grp['hiv_aids_deaths'],
            c=[cmap(norm(year))],
            marker=region_marker.get(region, 'o'),
            s=25,
            edgecolors='white',
            linewidths=0.2,
            label=region,
            zorder=3,
            alpha=0.85
        )

    reg_yr = yr_df[['gdp_per_capita', 'hiv_aids_deaths']].dropna()
    reg_yr = reg_yr[reg_yr['gdp_per_capita'] > 0]

    if len(reg_yr) >= 5:
        x_log = np.log(reg_yr['gdp_per_capita'].values)
        y_vals = np.log(reg_yr['hiv_aids_deaths'].values)
        slope, intercept, r_value, p_value, _ = scipy.stats.linregress(x_log,
            y_vals)
        r2 = r_value ** 2

        x_log_line = np.linspace(x_log.min(), x_log.max(), 200)
        ax.plot(
            np.exp(x_log_line),
            np.exp(slope * x_log_line + intercept),
            color='crimson',
            linewidth=1.2,
            linestyle='--',

```

```

        zorder=4
    )
    ax.set_title(f'{year} R2= $\{r2:.2f\}$ ', fontsize=9, pad=3)
else:
    ax.set_title(str(year), fontsize=9, pad=3)

ax.tick_params(labelsize=6)
ax.grid(True, which='both', linestyle='--', linewidth=0.3, alpha=0.5)

ax.set_xlabel('GDP/cap', fontsize=7)
ax.set_ylabel('Deaths', fontsize=7)
ax.set_xscale('log')
ax.set_yscale('log')
r2overTime.append((year, r_value))

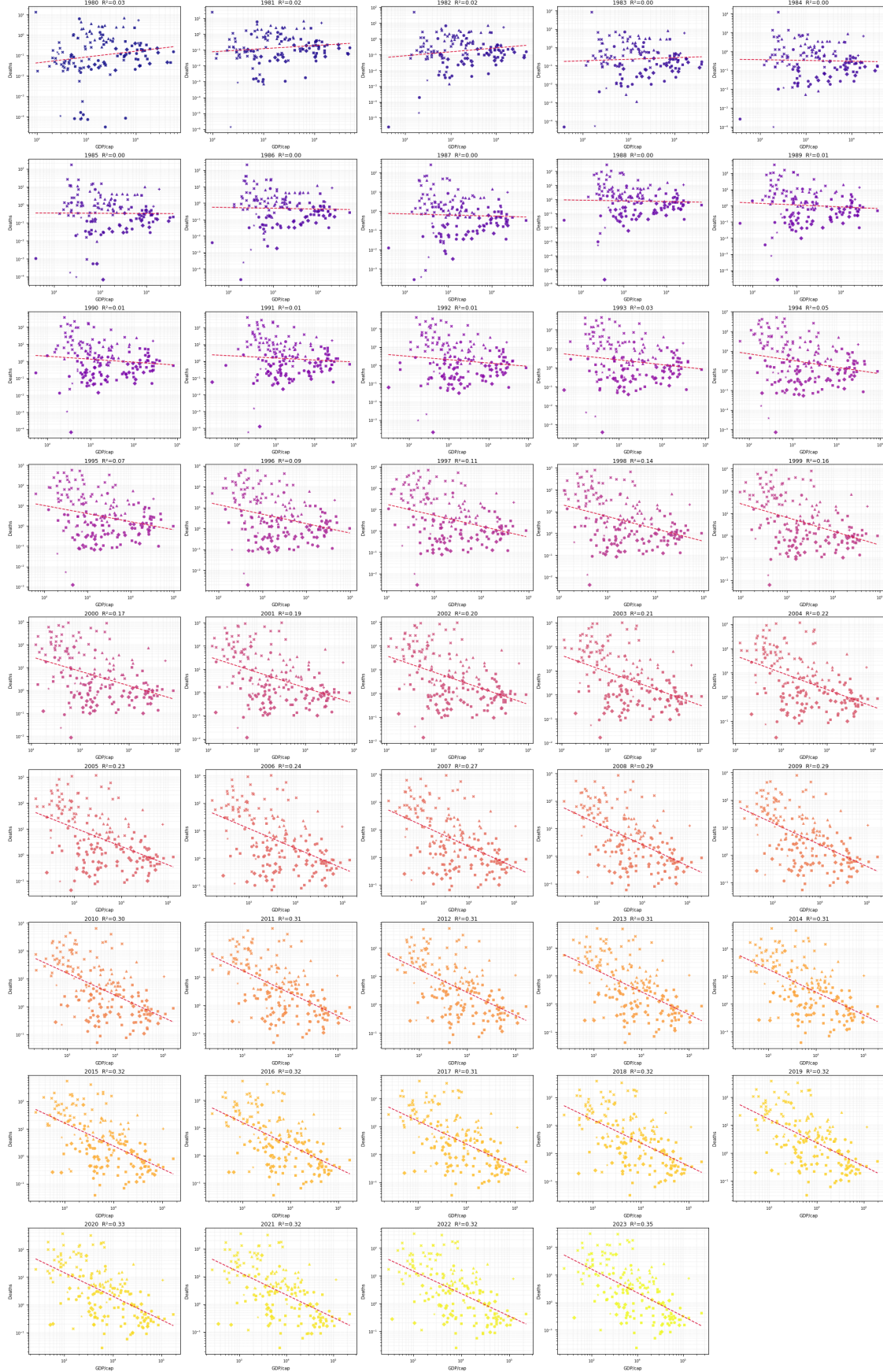
for ax in axes_flat[n_years:]:
    ax.set_visible(False)

handles, labels = axes_flat[0].get_legend_handles_labels()
fig.legend(
    handles, labels,
    title='WB Region',
    title_fontsize=8,
    fontsize=7,
    loc='lower center',
    ncol=4,
    bbox_to_anchor=(0.5, -0.03),
    framealpha=0.85
)

fig.suptitle('HIV/AIDS Deaths vs GDP per Capita - per year (log GDP vs regression)', fontsize=13)
plt.show()

```

HIV/AIDS Deaths vs GDP per Capita — per year (log GDP regression)



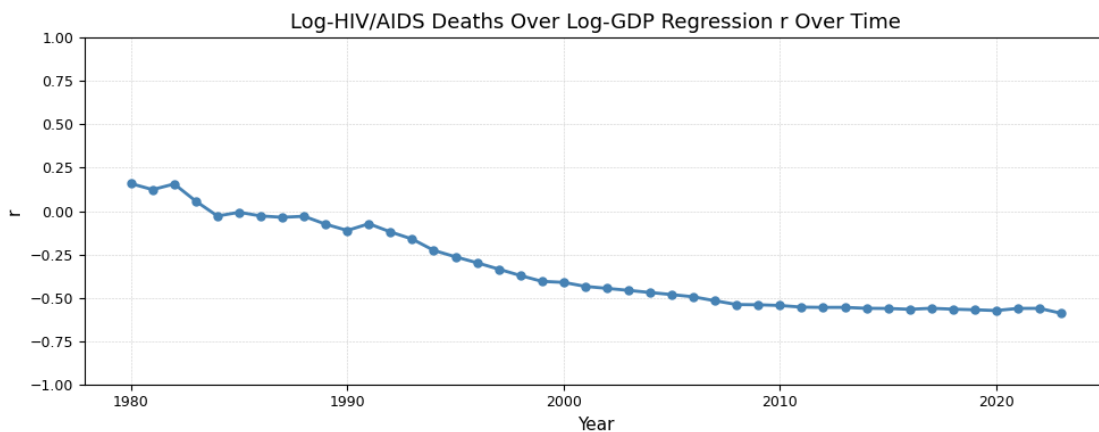
```
[44]: r2_df = pd.DataFrame(r2OverTime, columns=['year', 'r2'])

fig, ax = plt.subplots(figsize=(10, 4))

ax.plot(r2_df['year'], r2_df['r2'], color='steelblue', linewidth=2, marker='o',
        ↪markersize=5, zorder=3)

ax.set_xlabel('Year', fontsize=11)
ax.set_ylabel('r', fontsize=11)
ax.set_title('Log-HIV/AIDS Deaths Over Log-GDP Regression r Over Time',
        ↪fontsize=13)
ax.set_ylim(-1, 1)
ax.grid(True, which='both', linestyle='--', linewidth=0.4, alpha=0.6)
ax.tick_params(labelsize=9)

plt.tight_layout()
plt.show()
```



```
[45]: r2_df = pd.DataFrame(r2OverTime, columns=['year', 'r2'])

fig, ax = plt.subplots(figsize=(10, 4))

ax.plot(r2_df['year'], r2_df['r2']**2, color='steelblue', linewidth=2,
        ↪marker='o', markersize=5, zorder=3)

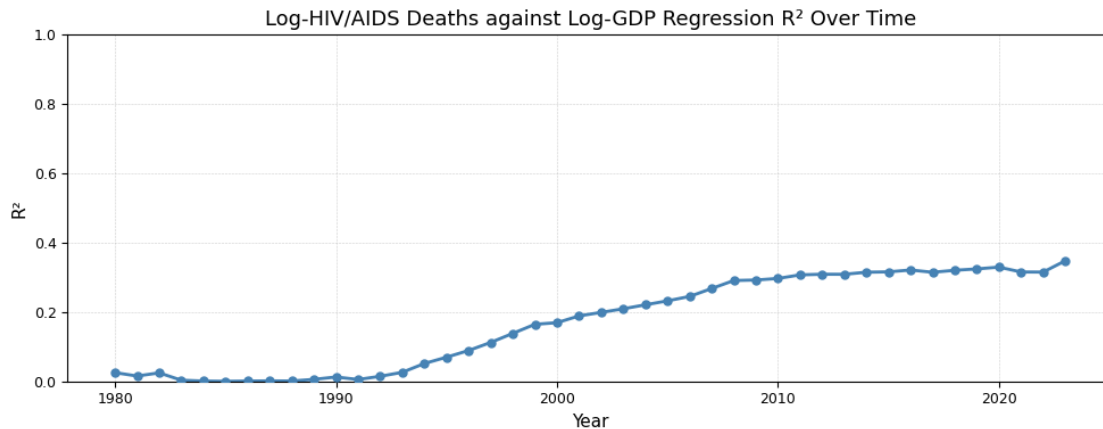
ax.set_xlabel('Year', fontsize=11)
ax.set_ylabel('R²', fontsize=11)
ax.set_title('Log-HIV/AIDS Deaths against Log-GDP Regression R² Over Time',
        ↪fontsize=13)
```

```

ax.set_ylim(0, 1)
ax.grid(True, which='both', linestyle='--', linewidth=0.4, alpha=0.6)
ax.tick_params(labelsize=9)

plt.tight_layout()
plt.show()

```



```

[46]: import pandas as pd

regions_lookup = pd.read_csv('world-regions-according-to-the-world-bank.csv')[
    ['Entity', 'World regions according to WB']
].drop_duplicates()

country_to_entity_map = {
    'Türkiye': 'Turkey',
    'Viet Nam': 'Vietnam',
    'Republic of Korea': 'South Korea',
    "Democratic People's Republic of Korea": 'North Korea',
    'Republic of Moldova': 'Moldova',
    'United Republic of Tanzania': 'Tanzania',
    'Venezuela (Bolivarian Republic of)': 'Venezuela',
    'Bolivia (Plurinational State of)': 'Bolivia',
    'Democratic Republic of the Congo': 'Democratic Republic of the
↪Congo',
    'Iran (Islamic Republic of)': 'Iran',
    'Syrian Arab Republic': 'Syria',
    'United States of America': 'United States',
    "Lao People's Democratic Republic": 'Lao PDR',
    'Brunei Darussalam': 'Brunei',
    "Côte d'Ivoire": 'Cote d'Ivoire',
    'Cabo Verde': 'Cape Verde',

```

```

}

plot_country_df = country_joined[
    (country_joined['sex'] == 'Both') &
    (country_joined['age'] == 'All ages') &
    (country_joined['population_group'] == 'All Population')
].copy()

plot_country_df['entity'] = plot_country_df['country'].
    ↪map(country_to_entity_map).fillna(plot_country_df['country'])

plot_country_df = plot_country_df.merge(
    regions_lookup,
    left_on='entity',
    right_on='Entity',
    how='left'
)

plot_country_df = plot_country_df.rename(columns={'World regions according to_
    ↪WB': 'wb_region'})

plot_country_df = plot_country_df[
    (plot_country_df['hiv_aids_deaths'] != 0.0)
].copy()

missing = plot_country_df[plot_country_df['wb_region'].isna()]['country'].
    ↪unique()

years = plot_country_df['year'].values
year_min, year_max = years.min(), years.max()

cmap = cm.plasma
norm = mcolors.Normalize(vmin=year_min, vmax=year_max)

regions = sorted(plot_country_df['wb_region'].dropna().unique())
markers = ['o', 's', '^', 'D', 'P', '*', 'X']
region_marker = {r: markers[i % len(markers)] for i, r in enumerate(regions)}

fig, ax = plt.subplots(figsize=(12, 8))

for region, grp in plot_country_df.groupby('wb_region'):
    grp = grp.sort_values('year')
    ax.scatter(
        grp['gdp_per_capita'],
        grp['hiv_aids_deaths'],
        c=grp['year'],
        cmap=cmap,

```

```

        norm=norm,
        marker=region_marker[region],
        s=50,
        edgecolors='white',
        linewidths=0.3,
        label=region,
        zorder=3
    )

reg_df = plot_country_df[['gdp_per_capita', 'year', 'hiv_aids_deaths']].dropna()
reg_df = reg_df[reg_df['gdp_per_capita'] > 0]

log_gdp = np.log(reg_df['gdp_per_capita'].values)
year_val = reg_df['year'].values
y_all = np.log(reg_df['hiv_aids_deaths'].values)

X = np.column_stack([
    np.ones(len(log_gdp)),
    log_gdp,
    year_val,
    log_gdp * year_val
])

beta, *_ = np.linalg.lstsq(X, y_all, rcond=None)

y_pred = X @ beta
ss_res = np.sum((y_all - y_pred) ** 2)
ss_tot = np.sum((y_all - y_all.mean()) ** 2)
r_squared = 1 - ss_res / ss_tot

print(f"2-variable OLS (log(GDP) + year + log(GDP)*year)")
print(f"  0 (intercept)      = {beta[0]:>12.4f}")
print(f"  1 (log GDP)           = {beta[1]:>12.4f}")
print(f"  2 (year)              = {beta[2]:>12.4f}")
print(f"  3 (log(GDP) * year)   = {beta[3]:>12.4f}")
print(f"  R²                    = {r_squared:.4f}")

x_log_line = np.linspace(log_gdp.min(), log_gdp.max(), 300)
x_line = np.exp(x_log_line)

sample_years = np.linspace(year_min, year_max, 4).astype(int)
year_colors = [cmap(norm(yr)) for yr in sample_years]

for yr, color in zip(sample_years, year_colors):
    y_line = np.exp(beta[0] + beta[1]*x_log_line + beta[2]*yr +
        ↪beta[3]*x_log_line*yr)
    ax.plot(

```

```

        x_line, y_line,
        color=color,
        linewidth=1.6,
        linestyle='--',
        zorder=4,
        label=f'Regression {yr}'
    )

sm = cm.ScalarMappable(cmap=cmap, norm=norm)
sm.set_array([])
cbar = fig.colorbar(sm, ax=ax, pad=0.02)
cbar.set_label('Year', fontsize=11)
cbar.set_ticks(np.arange(year_min, year_max + 1, 5))

ax.legend(
    title='WB Region / Regression',
    title_fontsize=9,
    fontsize=7,
    loc='lower right',
    framealpha=0.85
)

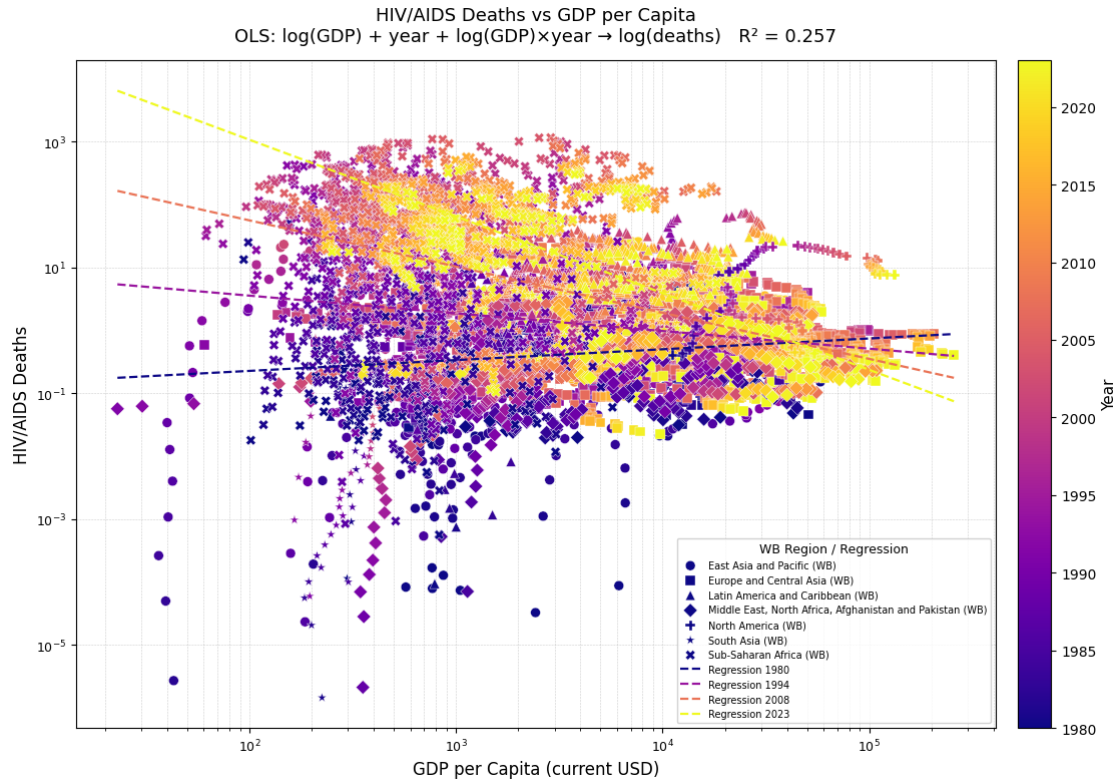
ax.set_xlabel('GDP per Capita (current USD)', fontsize=12)
ax.set_ylabel('HIV/AIDS Deaths', fontsize=12)
ax.set_title(
    f'HIV/AIDS Deaths vs GDP per Capita\n'
    f'OLS: log(GDP) + year + log(GDP)*year → log(deaths)    R² = {r_squared:.
    ↪3f}',
    fontsize=13, pad=14
)
ax.set_xscale('log')
ax.set_yscale('log')
ax.grid(True, which='both', linestyle='--', linewidth=0.4, alpha=0.6)
ax.tick_params(labelsize=9)

plt.tight_layout()
plt.show()

```

2-variable OLS (log(GDP) + year + log(GDP)*year)

0 (intercept)	=	-687.0186
1 (log GDP)	=	64.2440
2 (year)	=	0.3458
3 (log(GDP) × year)	=	-0.0324
R²	=	0.2573



```
[47]: regions_lookup = pd.read_csv('world-regions-according-to-the-world-bank.csv')[
    ['Entity', 'World regions according to WB']]
regions_lookup.drop_duplicates()

country_to_entity_map = {
    'Türkiye': 'Turkey',
    'Viet Nam': 'Vietnam',
    'Republic of Korea': 'South Korea',
    'Democratic People's Republic of Korea': 'North Korea',
    'Republic of Moldova': 'Moldova',
    'United Republic of Tanzania': 'Tanzania',
    'Venezuela (Bolivarian Republic of)': 'Venezuela',
    'Bolivia (Plurinational State of)': 'Bolivia',
    'Democratic Republic of the Congo': 'Democratic Republic of the Congo',
    'Iran (Islamic Republic of)': 'Iran',
    'Syrian Arab Republic': 'Syria',
    'United States of America': 'United States',
    'Lao People's Democratic Republic': 'Lao PDR',
    'Brunei Darussalam': 'Brunei',
    'Côte d'Ivoire': 'Cote d'Ivoire',
    'Cabo Verde': 'Cape Verde',
}
```

```

}

plot_country_df = country_joined[
    (country_joined['sex'] == 'Both') &
    (country_joined['age'] == 'All ages') &
    (country_joined['population_group'] == 'All Population')
].copy()

plot_country_df['entity'] = plot_country_df['country'].
    ↪map(country_to_entity_map).fillna(plot_country_df['country'])

plot_country_df = plot_country_df.merge(
    regions_lookup,
    left_on='entity',
    right_on='Entity',
    how='left'
)

plot_country_df = plot_country_df.rename(columns={'World regions according to_
    ↪WB': 'wb_region'})

plot_country_df = plot_country_df[
    (plot_country_df['hiv_aids_deaths'] != 0.0)
].copy()

missing = plot_country_df[plot_country_df['wb_region'].isna()]['country'].
    ↪unique()

regions = sorted(plot_country_df['wb_region'].dropna().unique())
markers = ['o', 's', '^', 'D', 'P', '*', 'X']
region_marker = {r: markers[i % len(markers)] for i, r in enumerate(regions)}

fig, ax = plt.subplots(figsize=(12, 8))

for region, grp in plot_country_df.groupby('wb_region'):
    grp = grp.sort_values('year')
    ax.scatter(
        grp['year'],
        grp['hiv_aids_deaths'],
        marker=region_marker[region],
        s=50,
        alpha=0.6,
        edgecolors='white',
        linewidths=0.3,
        label=region,
        zorder=3
    )

```

```

reg_df = plot_country_df[['year', 'hiv_aids_deaths']].dropna()

x_all = reg_df['year'].values
y_all = reg_df['hiv_aids_deaths'].values

slope, intercept, r_value, p_value, std_err = scipy.stats.linregress(x_all,
↪y_all)
r_squared = r_value ** 2

x_line = np.linspace(x_all.min(), x_all.max(), 300)
y_line = slope * x_line + intercept

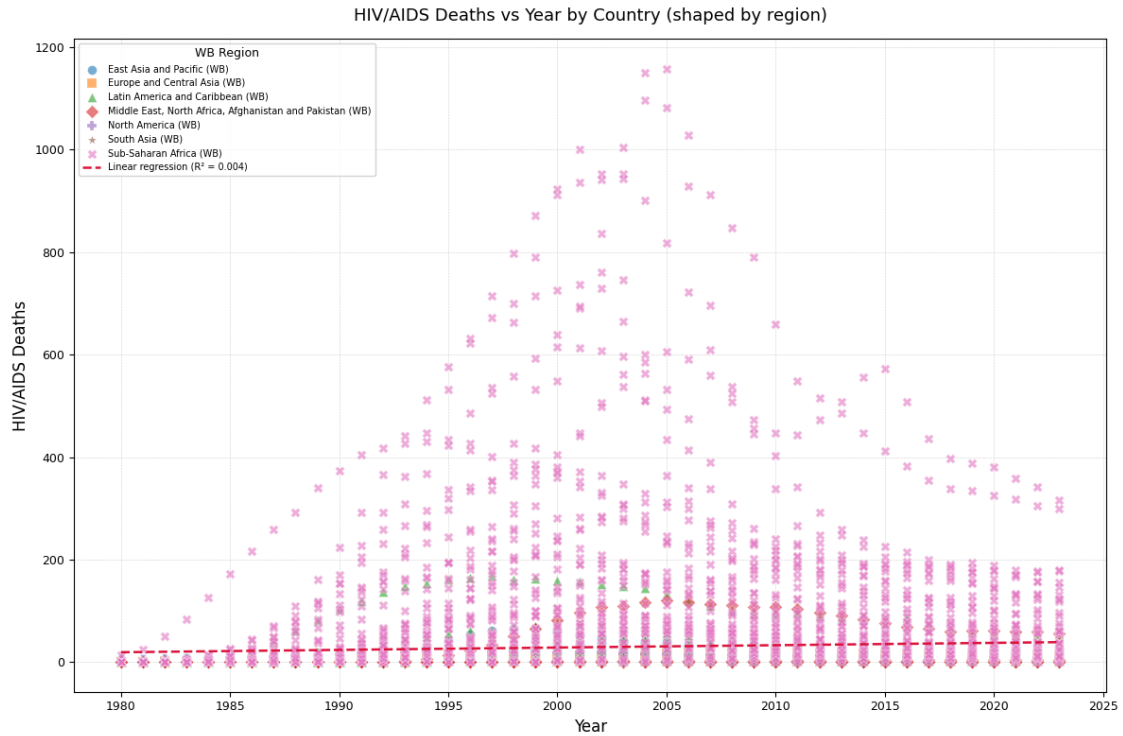
ax.plot(
    x_line, y_line,
    color='crimson',
    linewidth=1.8,
    linestyle='--',
    zorder=4,
    label=f'Linear regression ( $R^2 = \{r\_squared:.3f\}$ )'
)

ax.legend(
    title='WB Region',
    title_fontsize=9,
    fontsize=7,
    loc='upper left',
    framealpha=0.85
)

ax.set_xlabel('Year', fontsize=12)
ax.set_ylabel('HIV/AIDS Deaths', fontsize=12)
ax.set_title('HIV/AIDS Deaths vs Year by Country (shaped by region)',
↪fontsize=13, pad=14)
ax.grid(True, which='both', linestyle='--', linewidth=0.4, alpha=0.6)
ax.tick_params(labelsize=9)
ax.xaxis.set_major_locator(plt.MaxNLocator(integer=True))

plt.tight_layout()
plt.show()

```



```
[48]: target_years = [1981, 1996, 2010, 2023]

for year in target_years:
    yr_df = plot_country_df[plot_country_df['year'] == year]

    if yr_df.empty:
        print(f'No data for {year}')
        continue

    fig, ax = plt.subplots(figsize=(12, 8))

    for region, grp in yr_df.groupby('wb_region'):
        ax.scatter(
            grp['gdp_per_capita'],
            grp['hiv_aids_deaths'],
            c=[cmap(norm(year))],
            marker=region_marker.get(region, 'o'),
            s=60,
            edgecolors='white',
            linewidths=0.3,
            label=region,
            zorder=3,
            alpha=0.85
```

```

    )

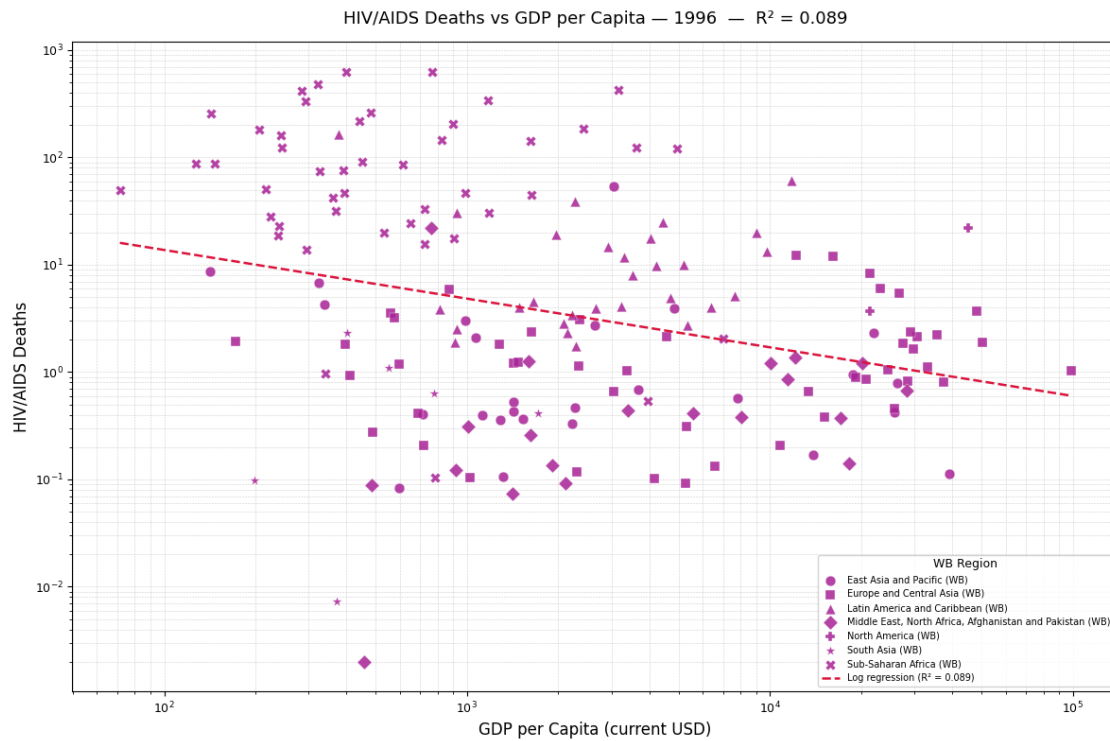
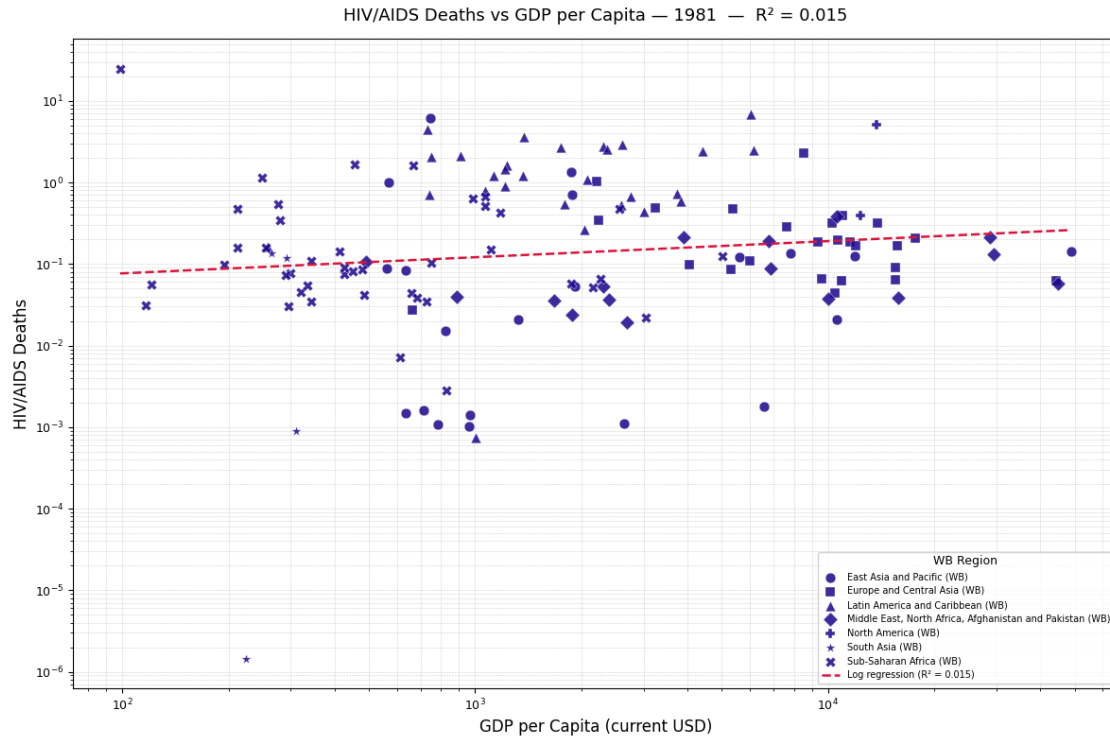
    reg_yr = yr_df[['gdp_per_capita', 'hiv_aids_deaths']].dropna()
    reg_yr = reg_yr[reg_yr['gdp_per_capita'] > 0]
    if len(reg_yr) >= 5:
        x_log = np.log(reg_yr['gdp_per_capita'].values)
        y_vals = np.log(reg_yr['hiv_aids_deaths'].values)
        slope, intercept, r_value, p_value, _ = scipy.stats.linregress(x_log,
↪y_vals)
        r2 = r_value ** 2
        x_log_line = np.linspace(x_log.min(), x_log.max(), 300)
        ax.plot(
            np.exp(x_log_line),
            np.exp(slope * x_log_line + intercept),
            color='crimson',
            linewidth=1.8,
            linestyle='--',
            zorder=4,
            label=f'Log regression ( $R^2 = {r2:.3f}$ )'
        )
        title_r2 = f' -  $R^2 = {r2:.3f}$ '
    else:
        title_r2 = ''

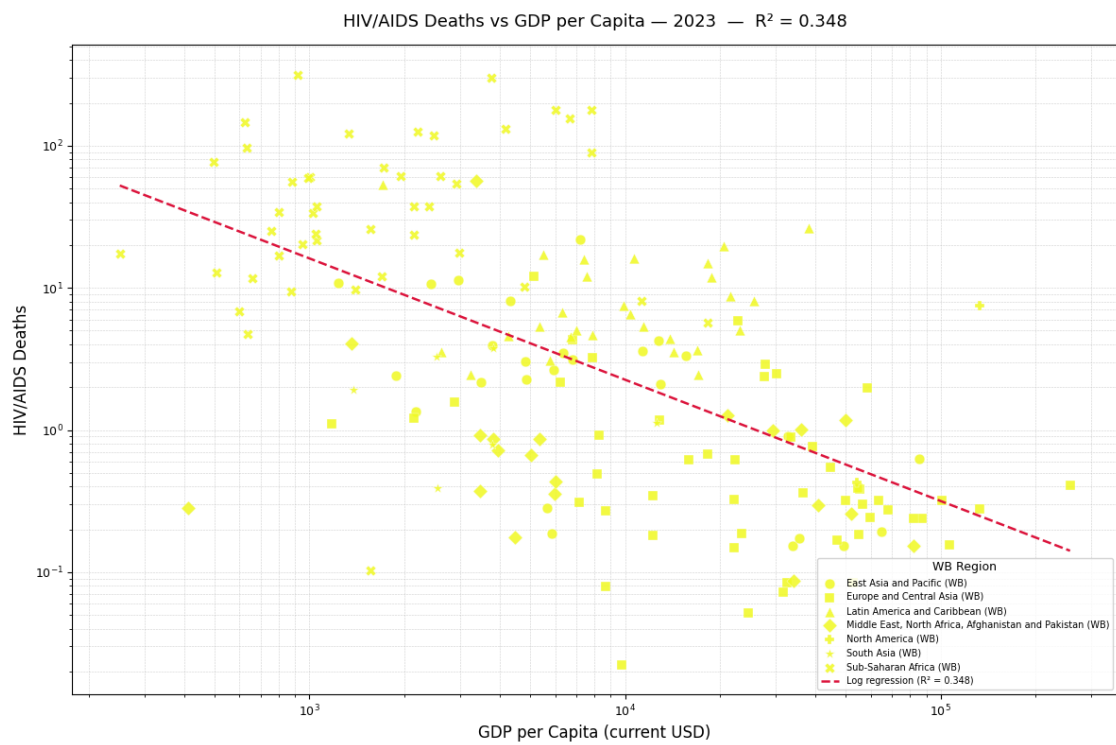
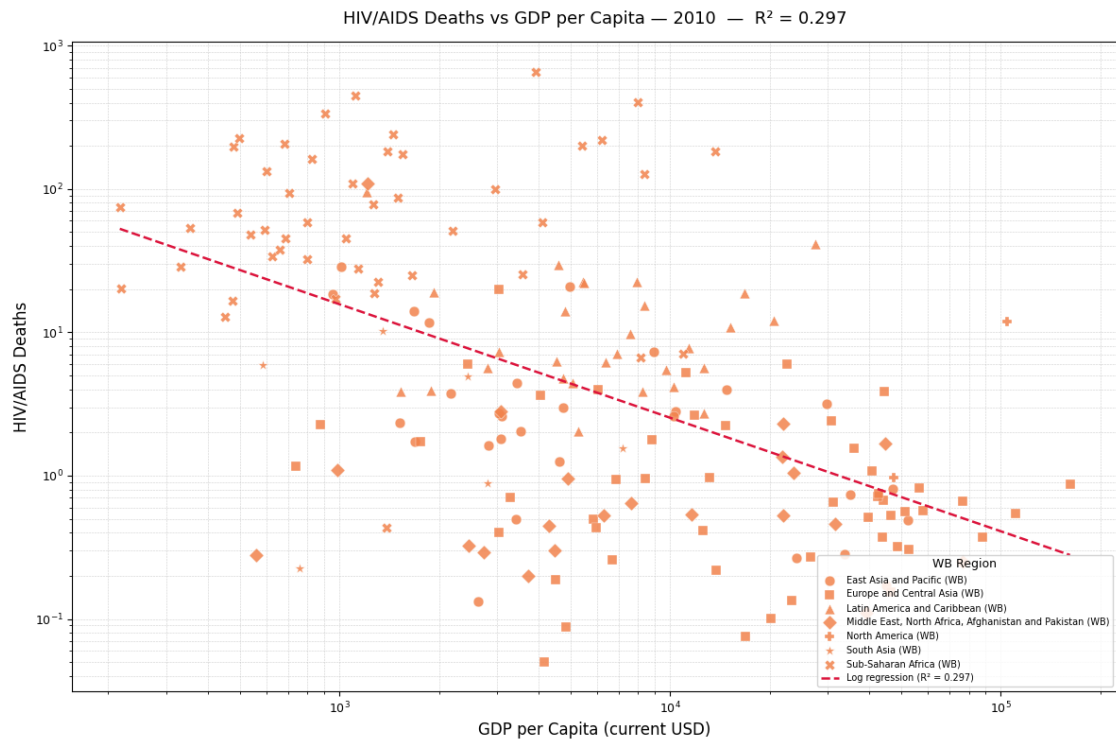
    ax.set_xscale('log')
    ax.set_yscale('log')
    ax.set_xlabel('GDP per Capita (current USD)', fontsize=12)
    ax.set_ylabel('HIV/AIDS Deaths', fontsize=12)
    ax.set_title(
        f'HIV/AIDS Deaths vs GDP per Capita - {year}{title_r2}',
        fontsize=13, pad=14
    )

    ax.grid(True, which='both', linestyle='--', linewidth=0.4, alpha=0.6)
    ax.tick_params(labelsize=9)
    ax.legend(
        title='WB Region',
        title_fontsize=9,
        fontsize=7,
        loc='lower right',
        framealpha=0.85
    )

    plt.tight_layout()
    plt.savefig(f'hiv_gdp_{year}.png', dpi=150, bbox_inches='tight')
    plt.show()

```





[]: